

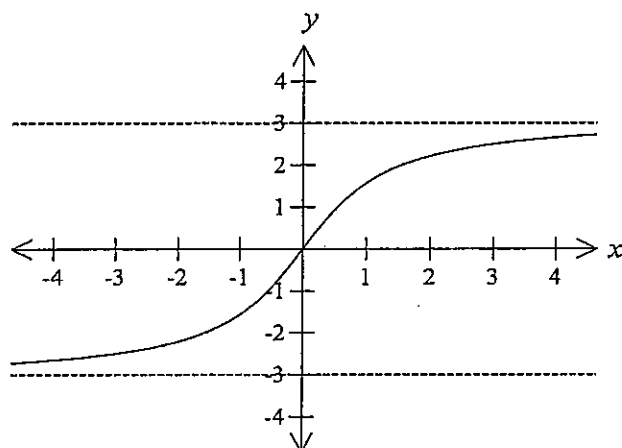
Name _____

Teacher _____

**GOSFORD HIGH SCHOOL****2012****HIGHER SCHOOL CERTIFICATE****ASSESSMENT TASK 2****MATHEMATICS – EXTENSION 2****Duration-** 90 minutes plus 5 minutes reading time

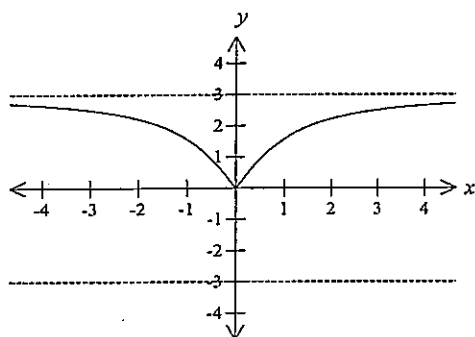
Multiple choice	8 questions worth 1 mark each. (Answer this section on the test paper)	/8
Graphs	2 questions worth 13 marks each. (Answer this section on your own paper. Start a new page for each question.)	/26
Polynomials	2 questions worth 13 marks each. (Answer this section on your own paper. Start a new page for each question.)	/26
TOTAL		/60

- 1 The diagram shows the graph of the function $y = f(x)$.

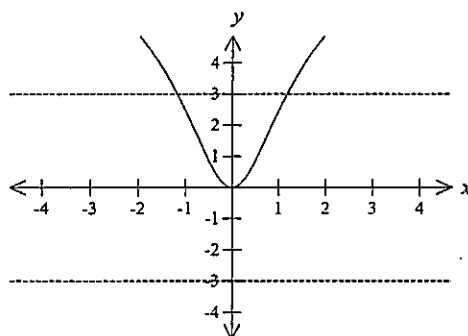


Which of the following is the graph of $y = \sqrt{f(x)}$?

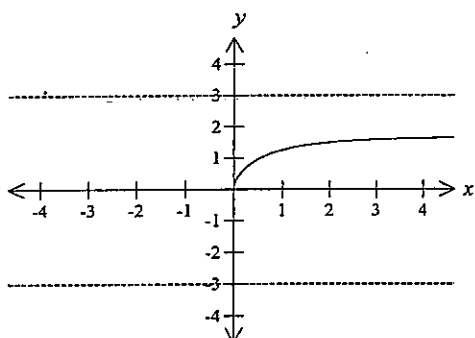
(A)



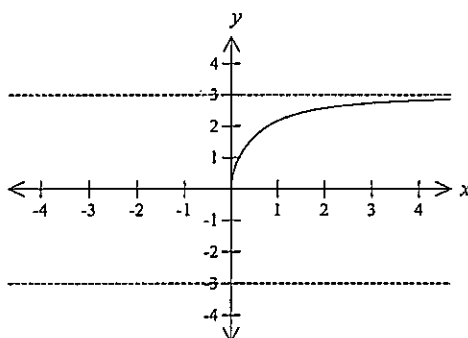
(B)



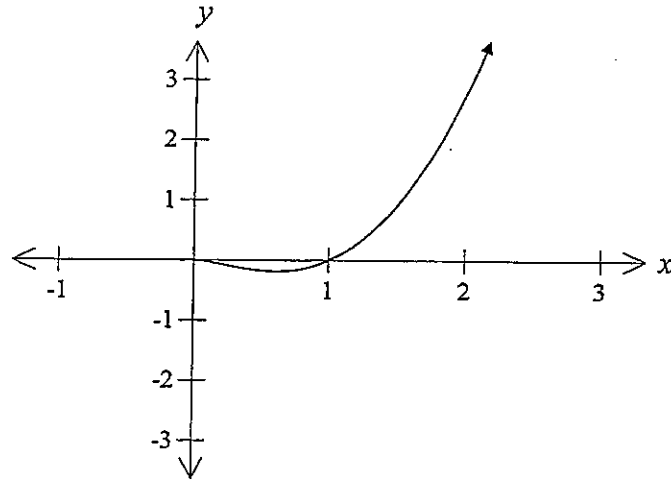
(C)



(D)

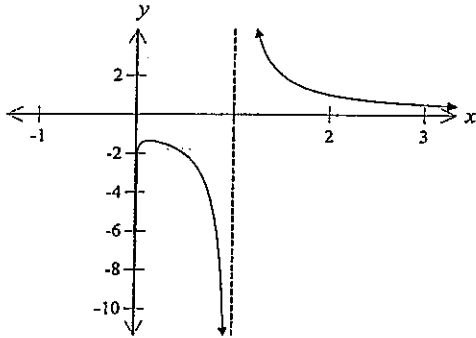


2 The diagram shows the graph of the function $y = f(x)$.

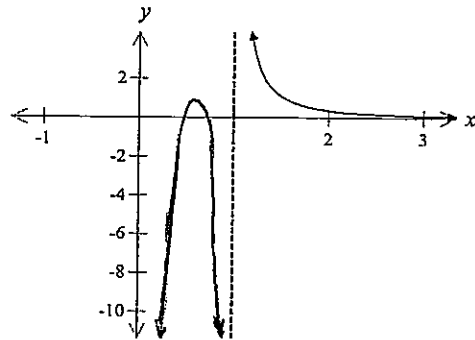


Which of the following is the graph of $y = \frac{1}{f(x)}$?

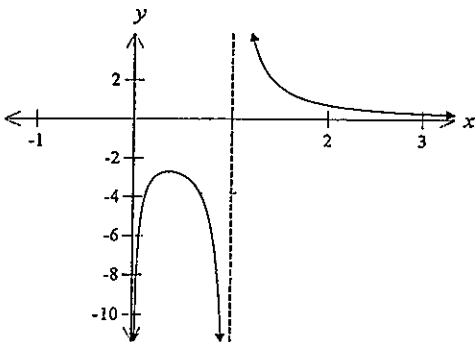
(A)



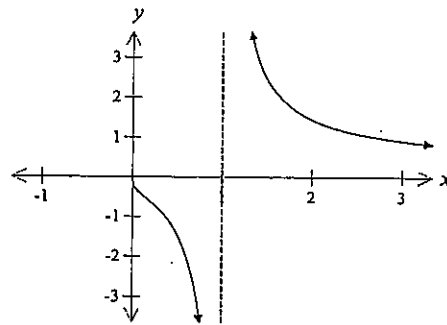
(B)



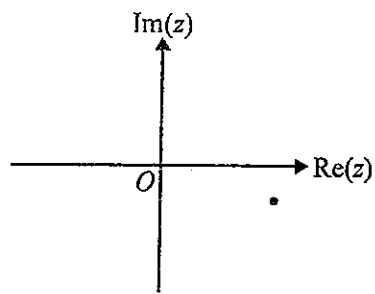
(C)



(D)



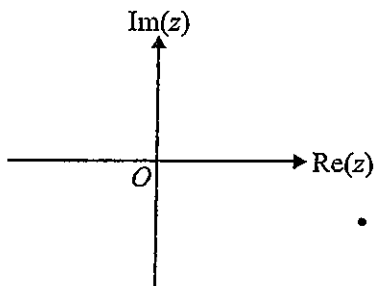
- 3 A certain complex number z , where $|z| > 1$, is represented by the point on the following argand diagram.



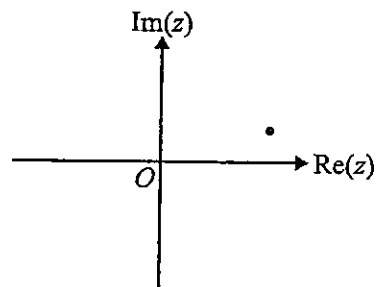
All axes below have the same scale as those in the diagram above.

The complex number $\frac{1}{\bar{z}}$ is best represented by

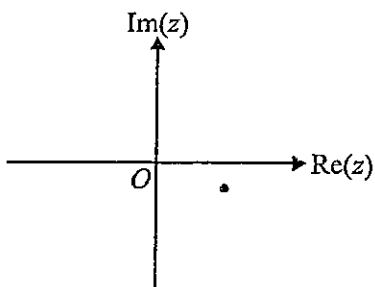
A.



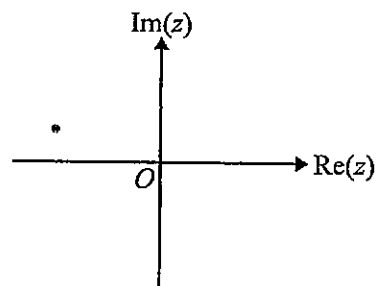
B.



C.



D.



- 4 Let $z = \text{cis}[150^\circ]$

The imaginary part of $z - i$ is

(A) $\frac{-i}{2}$

(B) $\frac{-1}{2}$

(C) $\frac{-\sqrt{3}}{2}$

(D) $\frac{-3i}{2}$

- 5 The polynomial $P(z)$ has real coefficients. Four of roots of the equation $P(z) = 0$ are $z = 0$, $z = 1 - 2i$, $z = 1 + 3i$, $z = 1 - 3i$ and $z = 3i$.

The minimum number of roots that the equation $P(z) = 0$ could have is

(A) 5

(B) 6

(C) 7

(D) 8

6 Given that $z = 4\text{cis}[120^\circ]$ it follows that the best answer to $\text{Arg}(z^5)$ in degrees is

(A) $(120^5)^\circ$

(B) 24°

(C) 600°

(D) -120°

7 The distance between the points z and $-\bar{z}$ in the complex plane is given by

(A) $2 \text{Re}(z)$

(B) $2 \text{Im}(z)$

(C) $2|z|$

(D) $2 \text{Re}(z) + 2 \text{Im}(z)$

6

8 The algebraic fraction $\frac{7}{(x-3)(x^2+4)}$ can be reduced to its partial fractions by equating it to

(A) $\frac{a}{(x-3)} + \frac{b}{(x^2+4)}$

(B) $\frac{a}{(x-3)} + \frac{b}{(x+2)(x-2)}$

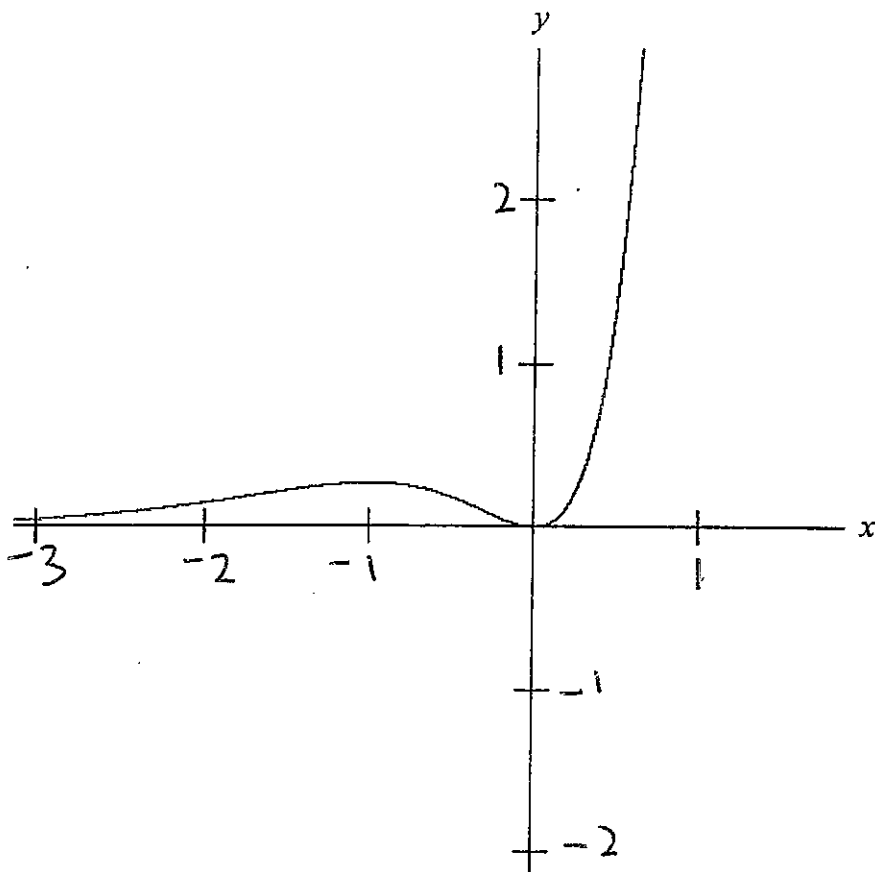
(C) $\frac{ax+b}{(x-3)} + \frac{c}{(x^2+4)}$

(D) $\frac{a}{(x-3)} + \frac{bx+c}{(x^2+4)}$

End of Multiple choice section (Total 8 marks)

Graphs (13 marks) Answer the rest of the exam on your own paper

Q1 The diagram below shows the graph of $y = f(x)$



Draw separate neat sketches of the following

- | | |
|-------------------------|---|
| i. $y = -f(x)$ | 1 |
| ii. $y = f(-x)$ | 2 |
| iii. $y = f x $ | 2 |
| iv. $y = \sqrt{f(x)}$ | 2 |
| v. $y = \frac{1}{f(x)}$ | 2 |
| vi. $y = f(x) $ | 2 |
| vii. $y^2 = f(x)$ | 2 |

Graphs (13 marks)

- Q2 (i) Show that $\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1}$ 4

Hence by adding ordinates or otherwise

sketch the graph of $y = \frac{x^2}{x+1}$

showing all important features.

- (ii) Consider the function $f(x) = \frac{3x}{(x-1)(4-x)}$

(a) Express $f(x)$ as partial fractions 2

(b) Find the x coordinates of any turning points 3

- (iii) On the same axes, neatly sketch the graphs of $y = (x+1)^2$

and $y = \frac{2}{x}$ (draw as dotted graphs for reference) By

multiplying ordinates or otherwise and using the 4

same axes, draw the graph of $y = \frac{2(x+1)^2}{x}$

(Make sure this graph is clearly distinguished from the reference graphs and not dotted)

Polynomials (13 marks)

- Q1 (a) The polynomial $P(x) = x^3 - 3x^2 + 4$ has a root of multiplicity 2. Find this root and fully factorise $P(x)$ 3
- (b) Factorise $x^3 - 4x^2 + 6x - 4$ into 3 linear factors 4
- (c) Form a new equation whose roots are reciprocals of the roots of $3x^3 + 4x^2 - 5x + 3 = 0$ 2
- (d) The equation $x^3 - x^2 - 3x + 2 = 0$ has roots α, β, γ
- (i) Use the value of $\alpha + \beta + \gamma$ to find the monic polynomial equation with roots $2\alpha + \beta + \gamma, \alpha + 2\beta + \gamma, \alpha + \beta + 2\gamma$ 2
- (e) α, β, γ are the roots of $x^3 + 2x^2 - 2x + 3 = 0$. Form the equation whose roots are $\alpha^2, \beta^2, \gamma^2$ 2

Polynomials (13marks)

Q2 (a) It is given that $3 - i$ is a root of $P(z) = z^3 + rz + 60$,
where r is a real number

(i) State why $3 + i$ is also a root of $P(z)$ 1

(ii) Factorise $P(z)$ over the real numbers 2

(b) By applying de Moivre's theorem and by expanding
 $(\cos\theta + i\sin\theta)^3$ obtain expressions for $\cos 3\theta$ in
terms of $\cos\theta$. 4

(c) (i) If $P(x)$ and $Q(x)$ have a common factor $(x - a)$,
show that $R(x) = P(x) - Q(x)$ 2
will have the same common factor

(ii) If $P(x) = 6x^3 + 7x^2 - x - 2$ and
 $Q(x) = 6x^3 - 5x^2 - 3x + 2$, find the two zeros 2
that $P(x)$ and $Q(x)$ have in common.

(d) When a polynomial $P(x)$ is divided by $(x - 3)$ and $(x - 4)$
the remainders are 5 and 12 respectively. Determine what the 2
remainder must be when $P(x)$ is divided by $(x - 3)(x - 4)$

End of examination

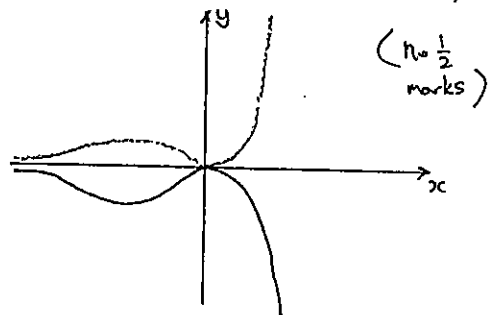
Please ensure

- 1. Your name or candidate number is on each section including the Question Sheet.**
- 2. Each of the 3 sections are stapled separately and collected for marking.**

2) 1/3 (-1 overall no ruler, untidy)
-1/2 no x,y labels

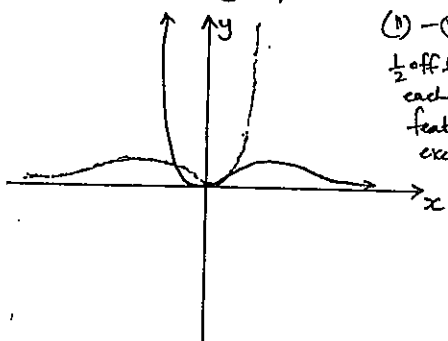
(1) $y = -f(x)$

(a reflection in the x axis)

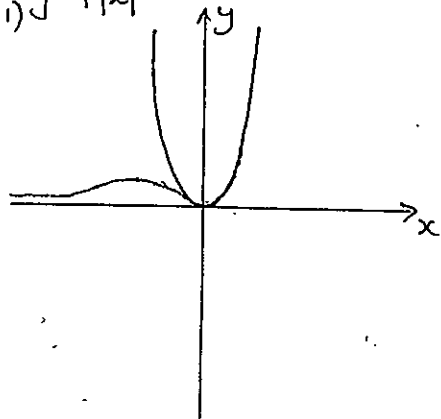


1) $y = f(-x)$

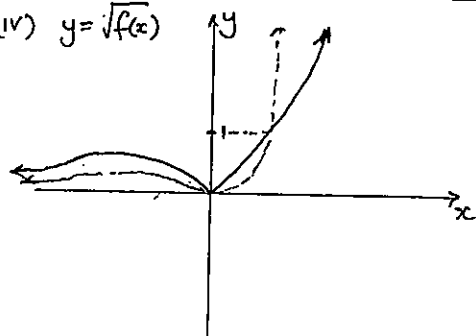
(1) - (vii)
1/2 off for each feature excluded.



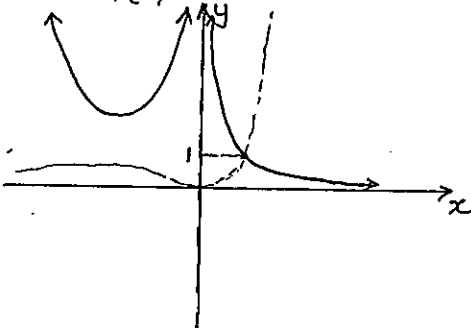
1) $y = f|x|$



(iv) $y = \sqrt{f(x)}$

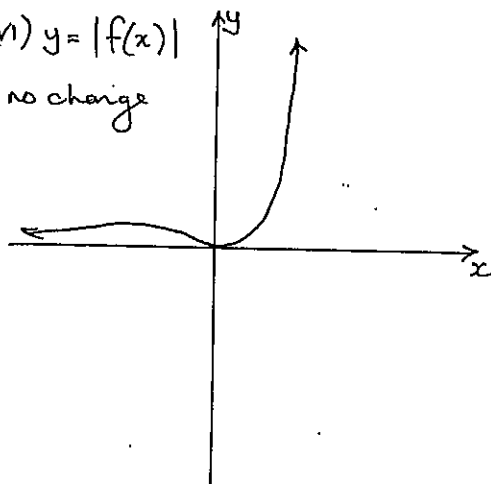


(v) $y = \frac{1}{f(x)}$



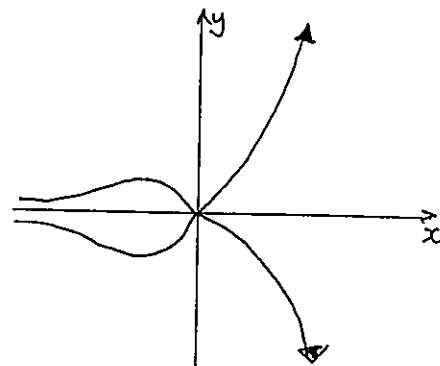
(vi) $y = |f(x)|$

no change



(vii) $y^2 = f(x)$

$y = \sqrt{f(x)}$ + reflection in x axis



Q2 (1) R.H.S. = $\frac{(x-1)(x+1)+1}{x+1}$

= $\frac{x^2-1+1}{x+1}$

= $\frac{x^2}{x+1}$ as req.

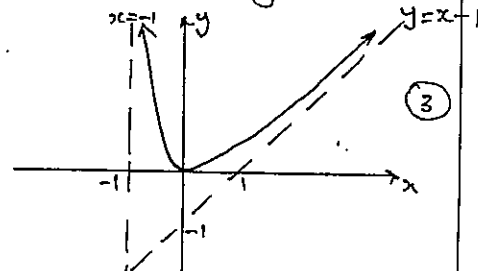
or by P.F.'s.

otherwise:

(1) V.A. $\Rightarrow x = -1$

H.A. $\Rightarrow y = x-1$ is an oblique asymptote

Intercepts $\Rightarrow x=0, y=0$



(11) let $y = \frac{3x}{(x-1)(4-x)}$

a) $\frac{3x}{(x-1)(4-x)} = \frac{a}{x-1} + \frac{b}{4-x}$

$3x = a(4-x) + b(x-1)$

(let $x=4$)

$12 = 3b$

$4 = b$

(let $x=1$)

$3 = 3a$

$1 = a$

$\therefore R.H.S. = \frac{1}{x-1} + \frac{4}{4-x}$

(1) b) stat pts occur when $y' = 0$

$y' = -1(x-1)^{-2} + -4(4-x)^{-2}$

= $-\frac{1}{(x-1)^2} + \frac{4}{(4-x)^2}$

($y' = 0$) $\frac{4}{(4-x)^2} = \frac{1}{(x-1)^2}$

$4(x-1)^2 = (4-x)^2$

$4x^2 - 8x + 4 = 16 - 8x + x^2$

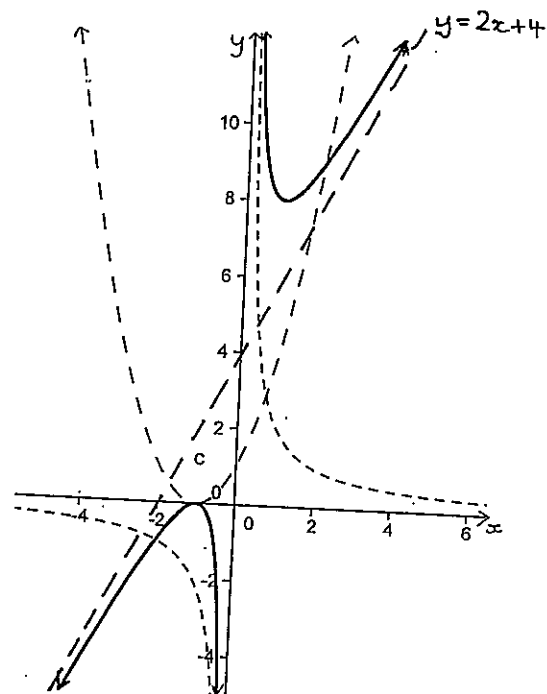
$3x^2 - 12 = 0$

$3(x-2)(x+2) = 0$

$\therefore x = \pm 2$

1 - each dotted
 (-1/2 no intercepts)
 1 - each arm
 (-1/2 no oblique)

Q2/
 (iii)



V.A $\Rightarrow x=0$ Intercepts $\Rightarrow y=0, x=-1$

$$HA \Rightarrow \frac{2x^2 + 4x + 2}{x} = 2x + 4 + \frac{2}{x}$$

$\therefore$ Oblique Asymptote $\Rightarrow y=2x+4$

MULTIPLE CHOICE ANSWERS

1/ C

2/ C

3/ C

4/ B

5/ C

6/ D

7/ A

8/ D

Polynomials

$$\begin{aligned} Q1(a) \quad P(x) &= x^3 - 3x^2 + 4 \\ P'(x) &= 3x^2 - 6x \\ &= 3x(x-2) \end{aligned}$$

$$\begin{aligned} x=0 & \text{ not a root} \quad \text{or} \quad x=2 \quad P(2)=0 \quad \therefore 2 \text{ is a root of multiplicity 2} \end{aligned}$$

$$\begin{aligned} \therefore P(x) &= (x-2)^2 Q(x) \\ &= (x-2)^2 (x+1) \quad (\text{by inspection}) \end{aligned}$$

$$(b) \quad x^3 - 4x^2 + 6x - 4$$

check $\pm 1, \pm 2, \pm 4$ $P(2)=0$

$$\begin{array}{r} x^2 - 2x + 2 \\ (x-2) \overline{) x^3 - 4x^2 + 6x - 4} \\ \underline{x^3 - 2x^2} \\ -2x^2 + 6x \\ \underline{-2x^2 + 4x} \\ 2x - 4 \\ \underline{2x - 4} \\ 0 \end{array}$$

$$\begin{aligned} x^3 - 4x^2 + 6x - 4 &= (x-2)(x^2 - 2x + 2) \\ &= (x-2)[(x-1)^2 + 1] \end{aligned}$$

$$= (x-2)[(x-1)^2 - i^2]$$

$$= (x-2)(x-1-i)(x-1+i)$$

Q1 Polynomials

$$(c) \quad 3x^3 + 4x^2 - 5x + 3 = 0$$

$$\frac{3}{x^3} + \frac{4}{x^2} - \frac{5}{x} + 3 = 0$$

$$\begin{aligned} & \text{multiply by } x^3 \\ 3 + 4x - 5x^2 + 3x^3 &= 0 \end{aligned}$$

$$\therefore 3x^3 - 5x^2 + 4x + 3 = 0 \text{ has reciprocal roots}$$

$$(d) \quad x^3 - x^2 - 3x + 2 = 0$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + \beta + \gamma = 1$$

$$\therefore 2\alpha + \beta + \gamma = \alpha + 1$$

$$\alpha + 2\beta + \gamma = \beta + 1$$

$$\alpha + \beta + 2\gamma = \gamma + 1$$

$$\therefore \text{replace } x \text{ with } (x-1)$$

$$(x-1)^3 - (x-1)^2 - 3(x-1) + 2 = 0 \quad \checkmark$$

$$\begin{aligned} x^3 - 3x^2 + 3x - 1 - x^2 + 2x - 1 - 3x + 3 + 2 &= 0 \\ x^3 - 4x^2 + 2x + 3 &= 0 \end{aligned} \quad \left. \begin{array}{l} \text{This step} \\ \text{not} \\ \text{necessary} \end{array} \right\}$$

$$(e) \quad x^3 - 2x^2 + x + 3 = 0$$

$$\text{let } x = \alpha^2 \quad \therefore \alpha = \sqrt{x} \text{ or } -\sqrt{x}$$

you only need one or the other not both.

$$\therefore x^2, \beta^2, \gamma^2 \text{ satisfies } (\sqrt{x})^3 - 2(\sqrt{x})^2 + (\sqrt{x}) + 3 = 0$$

$$x\sqrt{x} - 2x + \sqrt{x} + 3 = 0$$

$$x\sqrt{x} + \sqrt{x} = 2x - 3$$

$$\text{Squaring both sides} \quad x^3 + 2x^2 + x = 4x^2 - 12x + 9$$

$$x^3 - 2x^2 + 13x - 9 = 0$$

Polynomials

Q2/ (a) (i) $P(z)$ has real co-efficients so if $3-i$ is a root the conjugate $3+i$ is also a root

(ii) Let $Q(z) = z^2 - (3+i+3-i)z + (9+1)$
 $= z^2 - 6z + 10$

Now $P(z) = z^3 + rz + 60$

$\therefore z^3 + rz + 60 = (z^2 - 6z + 10)(z + 6)$

(b) $\cos 3\theta + i \sin 3\theta = (\cos \theta + i \sin \theta)^3$ (De Moivre's)
 $= \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3i^2 \sin^2 \theta \cos \theta + i^3 \sin^3 \theta$
 $= \cos^3 \theta - 3\sin^2 \theta \cos \theta + i(3\cos^2 \theta \sin \theta - \sin^3 \theta)$

Equating real

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3\sin^2 \theta \cos \theta \\ &= \cos^3 \theta - 3(1 - \cos^2 \theta) \cos \theta \\ &= \cos^3 \theta - 3\cos \theta + 3\cos^3 \theta \\ &= 4\cos^3 \theta - 3\cos \theta \end{aligned}$$

2(c)(i) let $P(x) = (x-a)A(x)$ and $Q(x) = (x-a)B(x)$

$$P(x) - Q(x) = R(x)$$

$$\begin{aligned} R(x) &= (x-a)A(x) - (x-a)B(x) \\ &= (x-a)(A(x) - B(x)) \end{aligned}$$

$\therefore R(x)$ has the same common factor as $P(x)$ and $Q(x)$

(ii) $6x^3 + 7x^2 - x - 2$ $6x^3 + 5x^2 + 3x - 2 = P(x) - Q(x)$
 $12x^2 + 2x - 4 = R(x)$

$$\begin{aligned} R(x) &= 2(6x^2 + x - 2) \\ &\quad \begin{array}{cc} 3x & 2 \\ 2x & -1 \end{array} \\ &= 2(3x+2)(2x-1) \end{aligned}$$

$\therefore$ Common zeros are $-\frac{2}{3}, \frac{1}{2}$

$$2(d) \quad P(x) = (x-3)(x-4) \cdot Q(x) + R(x) \\ = (x-3)(x-4) \cdot Q(x) + (ax+b)$$

$$P(3) = 5$$

$$\therefore 5 = 3a + b$$

$$P(4) = 12$$

$$\therefore 12 = 4a + b$$

$$4a + b = 12$$

$$3a + b = 5$$

subtract

$$a = 7$$

$$21 + b = 5$$

$$b = -16$$

$\therefore$ Remainder is $7x - 16$